

BloodHound Domain Graph Evidence Report

Internal Network & Active Directory Pentest Evidence Documentation

Field	Details
Test Area	BloodHound domain graph
Evidence Source	16-bloodhound-domain-overview-graph.png
Result Type	Domain relationships mapped
Primary Tool / Component	BloodHound / Neo4j
MITRE ATT&CK Mapping	T1069.002 - Domain Groups; T1087 - Account Discovery
Document Purpose	Professional GitHub-ready evidence report with redacted screenshot and interpretation.

This report documents one specific evidence row from the AD Network assessment. It is intentionally evidence-specific, not a generic attack tutorial. Sensitive values have been blurred or represented as redacted placeholders for safe portfolio publication.

1. Embedded Evidence Screenshot

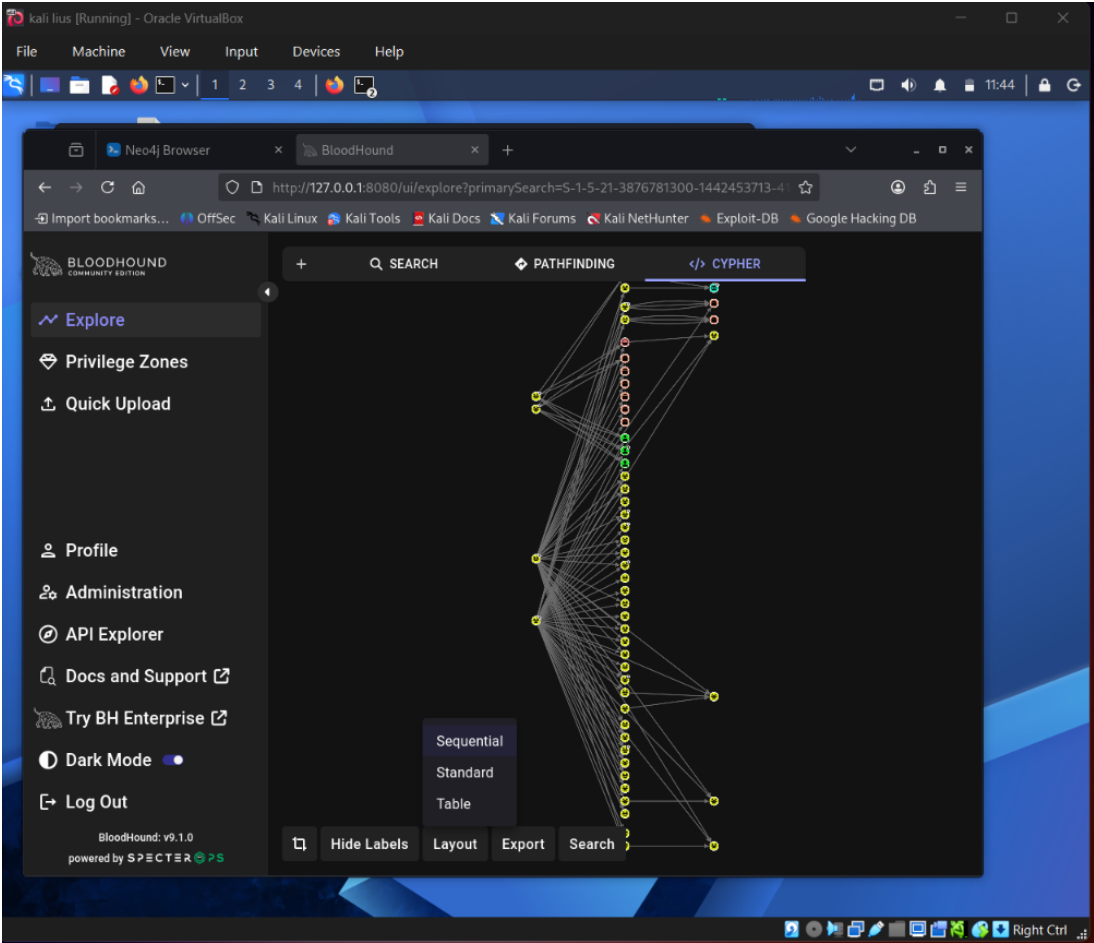


Figure 1 - Redacted evidence screenshot: 16-bloodhound-domain-overview-graph.png

What this screenshot proves

The screenshot shows a BloodHound domain overview graph where relationships between domain objects were imported and visualized.

Evidence Context

Area	Value
Testing Phase	Domain Relationship Mapping
Tool / Component	BloodHound / Neo4j
Objective	Document the overall AD graph generated from lab collection data and demonstrate relationship mapping across the domain.
Result Type	Domain relationships mapped

2. Evidence-Specific Methodology

This section explains how this evidence fits into the overall internal network and Active Directory attack path. The goal is to demonstrate professional testing logic, not simply show a terminal screenshot.

- Collect AD object and relationship data.
- Load the collection into BloodHound.
- Review domain objects, group relationships, and high-value principals.
- Use the graph to guide remediation planning.

Command / Workflow Used

```
bloodhound-python -d <DOMAIN> -u <USER> -p <REDACTED_PASSWORD> -dc <DC_IP> -c all --zip  
# BloodHound GUI: Overview / Domain graph queries
```

Sanitized Output Style

Domain graph loaded successfully. Relationships between users, computers, groups, and privileges are displayed for analysis.

Assessment Interpretation

The evidence supports the result type: Domain relationships mapped. In a professional assessment, this evidence would be paired with the exact testing scope, timestamps, target identifiers, and remediation ownership. For GitHub publication, the report intentionally avoids showing live secrets, passwords, hashes, or reusable credential material.

3. Risk Analysis

AD risk often comes from relationships rather than single vulnerabilities. A domain graph can reveal escalation chains hidden across multiple objects.

Why this matters in enterprise AD environments

- Active Directory attacks often chain multiple medium-risk issues into domain compromise.
- Credential material, even when not plaintext, can become dangerous if cracking, relaying, or alternate authentication is possible.
- Graph-based permissions and relationship analysis is necessary because AD risk is often hidden in group nesting and delegated rights.
- Evidence must be documented clearly so defenders can connect the technical finding to a specific control gap.

Recommended Remediation

- Use graph-based AD audits.
- Prioritize remediation of high-risk edges.
- Remove excessive group nesting.
- Limit privileged session exposure.
- Continuously monitor AD permission changes.

Detection and Monitoring Opportunities

- Correlate authentication events with unusual source hosts and off-hours activity.
- Monitor SMB, LDAP, Kerberos, and replication-related event patterns.
- Alert on abnormal use of administrative protocols and credential dumping indicators.
- Review domain privilege assignments and privileged group membership changes regularly.

4. GitHub Publication and Redaction Guidance

This report is prepared for safe GitHub portfolio publication. The embedded screenshot has been treated as evidence, but sensitive material must not be recoverable or readable.

Sensitive Item	Action for GitHub
NTLM / NetNTLMv2 hashes	Blur or remove completely. Never publish reusable hash material.
Kerberos TGS / krb5tgs values	Blur or replace with <REDACTED_TICKET>.
Plaintext passwords	Do not publish. Replace with <REDACTED_PASSWORD>.
DCSync / NTDS hashes	Treat as domain compromise material and fully redact.
Internal IPs / domain names	May remain for lab context, but redact if privacy is required.
Username / service accounts	Keep only if lab-only; otherwise redact or generalize.

Suggested GitHub Link

[Open Report](./reports/evidence/07-bloodhound-domain-graph-evidence-report.pdf)

Professional Note

This document should be stored with the AD Network module reports, preferably under 03-ad-network/reports/evidence/. It is designed to replace a plain screenshot filename in the Evidence-Based Testing Summary table with a professional clickable report.